

## Pre-Course Diagnostic: Algebra Skills Check

Before you begin

This calculus course assumes you're comfortable with a handful of algebra skills — not because you need to be an expert, but because Unit 1 (and the units right after it) will lean on these constantly, especially plugging expressions into functions and expanding things like  $(a + h)^2$ .

This is not a test to pass or fail. It's a map. Work through it without a calculator, check your answers against the key, and use the "What to review" guide at the end to figure out where to spend 15–20 minutes before you start Unit 1. If a topic feels shaky, that's completely normal — that's exactly what this diagnostic is for.

Take your time. There's no clock here.

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### Section 1: Order of Operations

1. Evaluate:  $3 + 4 \times 2^2$
2. Evaluate:  $(5 - 2) \times (7 - 4)$
3. Evaluate:  $12 \div 3 \times 2$

### Section 2: Working with Fractions

4. Compute:  $\frac{1}{3} + \frac{1}{4}$
5. Compute:  $\frac{2}{5} \times \frac{3}{4}$
6. Compute:  $\frac{3}{4} \div \frac{1}{2}$
7. Simplify:  $\frac{6}{8}$

### Section 3: Negative Numbers

8. Compute:  $-5 + 8$
9. Compute:  $-3 \times (-4)$
10. Compute:  $7 - (-2)$

### Section 4: Exponents

11. Evaluate:  $2^3$
12. Evaluate:  $(-3)^2$
13. Simplify (write as a single power):  $x^2 \cdot x^3$
14. Evaluate:  $5^0$

### Section 5: Expanding and Distributing

15. Expand:  $3(x + 4)$
16. Expand:  $(x + 2)^2$
17. Expand, in terms of  $x$  and  $h$ :  $(x + h)^2$
18. Expand:  $(a + 3)(a - 3)$

## Section 6: Factoring

19. Factor:  $x^2 + 5x + 6$
20. Factor:  $x^2 - 9$
21. Factor out the common factor:  $4x + 8$

## Section 7: Function Notation and Evaluation

22. If  $f(x) = x^2 + 1$ , find  $f(3)$ .
23. If  $f(x) = 2x - 5$ , find  $f(-2)$ .
24. If  $f(x) = x^2 + 3x$ , find  $f(a + h)$ . (Just substitute and expand — don't simplify further.)
25. If  $f(x) = x^2$ , find  $f(a + h) - f(a)$ .

## Section 8: Simplifying Algebraic Fractions

26. Simplify:  $\frac{2xh + h^2}{h}$  (factor the top, then cancel)
27. Combine into a single fraction:  $\frac{1}{x} - \frac{1}{2}$

## Section 9: Radicals (Square Roots)

28. Simplify:  $\sqrt{16}$
29. Simplify (as a simplified radical):  $\sqrt{50}$
30. Rationalize the denominator:  $\frac{1}{\sqrt{2}}$
31. Evaluate:  $\sqrt{9} + \sqrt{4}$

## Section 10: Solving Linear Equations

32. Solve for  $x$ :  $2x + 5 = 13$
33. Solve for  $x$ :  $3(x - 1) = 12$

## Section 11: Slope and Lines

34. Find the slope of the line through  $(1, 2)$  and  $(4, 11)$ .
35. Write the equation of a line with slope 2 that passes through  $(0, 5)$ .

## Section 12: The Coordinate Plane

36. Which quadrant is the point  $(-3, 5)$  in?

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When you're done, check your work against the answer key. Don't worry about getting everything perfect — just note which sections gave you the most trouble.